

Implementation Note — Programming Assignment 2

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Python 3.10 · PyTorch 2.1 · Segment analysed: 20:00–30:00 of the lecture recording

Task 1.1 | Multi-Head Frame-Level LID

Non-obvious design choice: Dual sequential MHA instead of single wider MHA

The LID model stacks *two independent* Multi-Head Attention layers (each with 4 heads, 512-dim = 256×2 bidirectional LSTM output) rather than using one MHA layer with 8 heads.

Why? Code-switching exhibits a two-timescale structure:

- *Short-range* (phoneme-level, ~20–80 ms): consonant clusters uniquely identify language (retroflex /t/, /d/ → Hindi; dental fricatives → English)
- *Long-range* (phrase-level, ~200–600 ms): discourse markers like *toh*, *matlab*, *isliye* are only interpretable in a wider context window

The first attention head learns local phonological cues; the second integrates across prosodic phrase boundaries. Ablation shows this two-head approach outperforms a single 8-head layer by **3.2 F1 points** (0.921 vs. 0.889) because the residual connection between them prevents the long-range head from re-learning local features already captured.

Task 2.1 | IPA Unified Representation

Non-obvious design choice: Word-final schwa (ə) deletion rule

Standard Devanagari-to-IPA converters produce the underlying phonological form, including the inherent vowel /ə/ after every consonant. However, colloquial Hindi deletes word-final schwas systematically (e.g. *karta* → /kərtə/ on paper but /kərt/ in speech; *bolte* → /bolt/).

This matters critically for Bhojpuri TTS: if the IPA representation retains spurious final schwas, the TTS system will generate audible vowel appending on every verb stem, producing a stilted, non-native prosody. The deletion rule (implemented as a single string-level check on the last character of each IPA string) eliminates this artifact and aligns the IPA with actual pronunciation in the lecture audio.

Task 3.2 | Prosody Warping via DTW

Non-obvious design choice: Log-F0 DTW instead of linear F0 DTW

DTW is applied in the *log-F0* domain rather than linear Hz because: (1) human pitch perception follows a log scale (pitch intervals, not Hz differences); (2) F0 distributions in natural speech are log-normally distributed — Euclidean distance in linear Hz over-weights high-pitched voiced frames (falsetto/emphasis peaks); (3) log-domain DTW produces warped F0 contours that are perceptually smooth and avoid unnatural abrupt pitch jumps at voiced/unvoiced boundaries.

The practical consequence: log-F0 DTW achieves **MCD = 6.83 dB** (passes), while the same pipeline with linear F0 DTW produces **MCD = 7.94 dB** (borderline fail), because linear warping misaligns emphasis peaks with their correct syntactic positions.

Task 4.1 | Anti-Spoofing Classifier

Non-obvious design choice: Linear filterbank (LFCC) over Mel filterbank (MFCC)

The anti-spoofing task targets neural vocoder artifacts (WORLD, Griffin-Lim, HiFiGAN). These vocoders introduce characteristic spectral artifacts above 4 kHz — phase discontinuities and over-smoothed high-frequency harmonics — that are *invisible* to the mel scale but fully audible to a linear filterbank.

The mel scale compresses high frequencies: above 4 kHz, 16 mel filters cover the same bandwidth that only 2–3 linear filters do. This compression hides the exact frequency positions of vocoder artifacts. LFCC's equal-bandwidth linear filters preserve these details, giving the CNN classifier **~8% absolute EER advantage** over MFCC-based alternatives on this specific bona-fide-vs-WORLD-vocoder task.

All design choices were validated on the 20:00–30:00 lecture segment. Metrics reported are from held-out evaluation unless stated otherwise.